

Mathieu Pont

Post-doctoral Researcher

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 0000-0002-0037-0314  RPTU Kaiserslautern-Landau, Scientific Visualization Lab, Germany.

Work Experience

Oct. 2024 – Ongoing	Post-doctoral Position <i>RPTU Kaiserslautern-Landau, Germany (Scientific Visualization Lab).</i>
Dec. 2023 – Sep. 2024	Post-doctoral Position <i>CNRS and Sorbonne Université (LIP6).</i>
2020 (6 months)	Master 2 Research Intern <i>CNRS and Sorbonne Université (LIP6).</i> Title: <i>Topologically Discriminant Metric.</i> Advisor: <i>Julien Tierny.</i>
2019 (3 months)	Master 1 Research Intern <i>Paris Descartes University (LIPADE).</i> Title: <i>Biomedical Corpus Analysis.</i> Advisor: <i>Séverine Affeldt.</i>
2018 (3 months)	Bachelor Research Intern <i>Toulouse Paul Sabatier University (IRIT).</i> Title: <i>Comparison of Deep Reinforcement Learning methods with an existing Multi-Agent System.</i> Advisors: <i>Frédéric Migeon and Jérôme Mengin.</i>
2016 (3 months)	DUT Research Intern <i>ISAE-Supaero.</i> Title: <i>Server Room Thermal Monitoring and Evaluation of EV3 Robotic Kit.</i> Advisors: <i>Régine Leconte and Jean-François Dassieu.</i>

Education

Oct. 2020 – Nov. 2023	Ph.D. in Computer Science <i>CNRS and Sorbonne Université (LIP6).</i> Title: <i>Analysis of Ensembles of Topological Descriptors.</i> Advisor: <i>Julien Tierny.</i>
2018 – 2020	Master's Degree in Computer Science <i>"Machine Learning for Data Science" track of Paris Descartes University.</i> Rank: <i>1 / 38 (S4) ; 1 / 37 (S3) ; 1 / 33 (S2) and 3 / 33 (S1)</i>
2016 – 2018	Bachelor's Degree in Computer Science <i>Toulouse Paul Sabatier University.</i> Rank: <i>4 / 152</i>
2014 – 2016	DUT GEII (Electrical and Computer Science Engineering) <i>Toulouse Paul Sabatier University.</i>

Awards

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| 2023 | <ul style="list-style-type: none">• Best Paper Honorable Mention Award at IEEE VIS 2023
<i>For the paper: "Merge Tree Geodesics and Barycenters with Path Mappings"</i>• Best Paper and Presentation Award at CORESA 2023
<i>For the talk: "Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance)"</i> |
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Research

Supervision

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| Ph.D. Student | <ul style="list-style-type: none">• Michelle Ditz (Jun. 2026 – Ongoing), co-supervised with Christoph Garth |
| Master Student | <ul style="list-style-type: none">• Michelle Ditz (Sep. 2025 – Feb. 2026)
Title: <i>Wasserstein Branch Mapping Distance for Merge Trees through Complete Branch Decompositions.</i> |
| Bachelor Student | <ul style="list-style-type: none">• Bastian Schmitt (Jun. 2026 – Nov. 2026)
Title: <i>Quantitative Assessment of Shape Similarity Techniques for Discrete Surface Representations.</i> |

Thesis

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| 2023 | <ul style="list-style-type: none">• Analysis of Ensembles of Topological Descriptors
<u>Mathieu Pont</u>
Ph.D. thesis in Computer Science
Committee: <i>Gabriel Peyré (President), David Coeurjolly (Reviewer), Vijay Natarajan (Reviewer), Elsa Cazelles (Examiner), Stanley Durrleman (Examiner), Roland Kwitt (Examiner), Katharine Turner (Examiner), Julien Tierny (Advisor)</i> |
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Publications

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| 2025 | <ul style="list-style-type: none">• Region-Aware Wasserstein Distances of Persistence Diagrams and Merge Trees
<u>Mathieu Pont</u> and Christoph Garth
<i>IEEE Transactions on Visualization and Computer Graphics.</i> |
| 2024 | <ul style="list-style-type: none">• A Practical Solver for Scalar Data Topological Simplification
Mohamed Kissi, <u>Mathieu Pont</u>, Joshua A. Levine and Julien Tierny
<i>IEEE Transactions on Visualization and Computer Graphics.</i>
Proc. of IEEE VIS 2024. |
| 2023 | <ul style="list-style-type: none">• Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)
<u>Mathieu Pont</u> and Julien Tierny
<i>IEEE Transactions on Visualization and Computer Graphics.</i>
Presented at IEEE VIS 2024.• Merge Tree Geodesics and Barycenters with Path Mappings
Florian Wetzels, <u>Mathieu Pont</u>, Julien Tierny and Christoph Garth
<i>IEEE Transactions on Visualization and Computer Graphics.</i>
Proc. of IEEE VIS 2023.
Best Paper Honorable Mention Award |

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| 2022 | <ul style="list-style-type: none"> • Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams)
 <u>Mathieu Pont</u>, Jules Vidal and Julien Tierny
 <i>IEEE Transactions on Visualization and Computer Graphics</i>.
 Presented at IEEE VIS 2023. |
| 2021 | <ul style="list-style-type: none"> • Wasserstein Distances, Geodesics and Barycenters of Merge Trees
 <u>Mathieu Pont</u>, Jules Vidal, Julie Delon and Julien Tierny
 <i>IEEE Transactions on Visualization and Computer Graphics</i>.
 Proc. of IEEE VIS 2021. |

Technical Reports

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| 2025 | <ul style="list-style-type: none"> • Hands-On TTK Tutorial: Solving Practical Problems in Small Groups
 Jonas Lukasczyk, Christoph Garth, Michael Will, <u>Mathieu Pont</u> and Julien Tierny
 <i>IEEE VIS Tutorials 2025</i>. |
| 2023 | <ul style="list-style-type: none"> • A Hands-on TTK Tutorial for Absolute Beginners
 Christoph Garth, Robin Maack, <u>Mathieu Pont</u>, Julien Tierny, Bei Wang, Florian Wetzels, Michael Will
 <i>IEEE VIS Tutorials 2023</i>. |
| 2022 | <ul style="list-style-type: none"> • Topological Analysis of Ensemble Scalar Data with TTK, A Sequel
 Christoph Garth, Charles Gueunet, Pierre Guillou, Federico Iuricich, Joshua Levine, Jonas Lukasczyk, <u>Mathieu Pont</u>, Julien Tierny, Jules Vidal, Bei Wang and Florian Wetzels
 <i>IEEE VIS Tutorials 2022</i>. |

Professional Service

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| Program
Committee | <ul style="list-style-type: none"> • IEEE VIS Short Papers
 2025, 2026 |
| Reviewer | <ul style="list-style-type: none"> • IEEE VIS
 2026 • EuroVis
 2026 • La Matematica
 2024 • IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
 2023 |

Talks

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| 2025 | <ul style="list-style-type: none"> • Tutorial: Merge Tree Computation and Feature Tracking in TTK
 Nov. 3rd, <i>IEEE VIS</i> |
| 2024 | <ul style="list-style-type: none"> • Poster: Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)
 Nov. 21st, <i>CORESA</i> • Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)
 Oct. 17th, <i>IEEE VIS</i> |

2024 (continued)	• Variability Analysis of Ensembles of Topological Descriptors <i>Jul. 1st, RPTU Kaiserslautern-Landau Invited Talk</i> • Auto-Encodeurs de Wasserstein d'Arbres de Fusion (et de Diagrammes de Persistance) <i>Jun. 18th, Journée Visu</i> • Auto-Encodeurs de Wasserstein d'Arbres de Fusion (et de Diagrammes de Persistance) <i>May 30th, Journée APR</i>
2023	• Analysis of Ensembles of Topological Descriptors <i>Dec. 1st, Ph.D. Defense</i> • Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams) <i>Oct. 26th, IEEE VIS</i> • Tutorial: Wasserstein Distances between Persistence Diagrams in TTK <i>Oct. 22nd, IEEE VIS</i> • Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams) <i>Oct. 16th, Pre-VIS Day</i> • Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance) <i>Jun. 23rd, Journée APR</i> • Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance) <i>Jun. 22nd, Journée Visu</i> • Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance) <i>Jun. 8th, CORESA – Best Paper and Presentation Award!</i>
2022	• Distances de Wasserstein, Géodésiques et Barycentres d'Arbres de Fusion <i>Nov. 25th, JFIG</i> • Tutorial: Wasserstein Distances, Barycenters and Clusters of Merge Trees in TTK <i>Oct. 17th, IEEE VIS, Recorded Talk</i> • Distances de Wasserstein, Géodésiques et Barycentres d'Arbres de Fusion <i>Jun. 28th, Journée Visu</i>
2021	• Wasserstein Distances, Geodesics and Barycenters of Merge Trees <i>Oct. 28th, IEEE VIS, Recorded Talk</i>

Teaching Experience

2024 – 2025	• RPTU Kaiserslautern-Landau – Scientific Visualization <i>~ 2h in Master 2 using C++ and Paraview</i>
2022 – 2023	• Sorbonne Université – Introduction to Programming 1 <i>~ 40h in Bachelor 1 using Python</i> • Sorbonne Université – Data Structures <i>~ 20h in Bachelor 2 using C</i>
2021 – 2022	• Sorbonne Université – Introduction to Programming 1 <i>~ 40h in Bachelor 1 using Python</i> • Sorbonne Université – Introduction to Scientific Visualization <i>~ 20h in Master 2 using C++ and ParaView</i>
2020 – 2021	• Sorbonne Université – Introduction to Scientific Visualization <i>~ 20h in Master 2 using C++ and ParaView</i> • Sorbonne Université – Introduction to Programming 2 <i>~ 40h in Bachelor 1 using C</i>